

COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 6

5 Computation Theory (amp12)

Primitive and
partial recursive
functions

Given a partial function $f : \mathbb{N} \rightarrow \mathbb{N}$, let $D(f)$ denote the set of natural numbers at which f is defined: $D(f) = \{x \in \mathbb{N} \mid f(x) \downarrow\}$; and let $I(f)$ be the set of natural numbers that are values of f where it is defined: $I(f) = \{y \in \mathbb{N} \mid \text{for some } x, f(x) = y\}$.

Prove or disprove the following statements, clearly stating any results about register machine computable functions and partial recursive functions that you use.

Answer: The following facts are used in the answers below:

- (A) The set of partial recursive functions is equal to the set of Register Machine (RM) computable partial functions.
 - (B) There is a universal RM that can decode numbers as RM programs and run them.
 - (C) There is a RM computable bijection $(m, n) \mapsto \langle m, n \rangle$ between $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$.
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- (a) Every subset $S \subseteq \mathbb{N}$ is equal to $I(f)$ for some register machine computable partial function f . [4 marks]

Answer: False. Since one can encode RM programs as numbers, there are only countably many of them and hence only countably many subsets of $\mathbb{N}$ of the form $I(f)$ for some RM computable f . But (by Cantor's Diagonal Argument), there are uncountably many subsets of $\mathbb{N}$.

- (b) If f is register machine computable, then $I(f)$ is equal to $D(g)$ for some partial recursive function g . [7 marks]

Answer: True. By fact (A) it suffices to find a RM computable partial function g such that $I(f) = D(g)$. Let M be a RM computing f . Using fact (B), from the program of M one can construct a new RM M' to carry out the following computation:

given input x , for each number $n = 0, 1, 2, \dots$ in turn, decode n as a pair $n = \langle k, m \rangle$ (using fact (C)), run the program of M for k steps starting with input m and if it has halted with output x , then halt with output 1, else repeat for the next number $n + 1$.

The function g computed by M' satisfies for all $x \in \mathbb{N}$ that $g(x) = 1$ iff $x \in I(f)$ and is undefined otherwise. Thus $I(f) = D(g)$.

- (c) If f is register machine computable, then $D(f)$ is equal to $I(g)$ for some *total* recursive function g . [4 marks]

Answer: False. If f is the totally undefined function it is clearly RM computable and satisfies $D(f) = \emptyset$. But for any *total* function $g : \mathbb{N} \rightarrow \mathbb{N}$, the set $I(g)$ is non-empty (for example it contains $g(0)$).

- (d) If g is a partial recursive function, then $D(g)$ is equal to $I(f)$ for some register machine computable partial function f . [5 marks]

Answer: True. By fact (A), g is computable by some RM M . Using fact (B) we can construct a new register machine M' that

stores a copy of its input x , runs the program of M starting with x and if that halts, copies x to the output register and halts.

The function f computed by M' satisfies for all $x \in \mathbb{N}$ that $f(x) = x$ iff $x \in D(g)$ and is undefined otherwise. Thus $D(g) = I(f)$.
